

**Higher Diploma in Science in Data
Analytics for Business
Level 8**

Programme Handbook

Awarding Body

Quality and Qualifications Ireland

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Section 1: Introduction

1.1 Programme Overview

The Higher Diploma in Science in Data Analytics for Business is designed for learners seeking to develop their knowledge, skills and competence in the area of Data Analytics. This programme is designed to produce graduates with the attributes required of a Data Analyst today, and with the ability to continue to develop knowledge, skill and competence to remain competitive and employable in a rapidly changing and expanding sector of the economy.

Learners will be provided with the opportunity to assimilate knowledge within an industry focused learning environment. This focus is maintained through the use of practical sessions in labs and workshops. Graduates will be qualified to assume entry level industry roles and/or to further advance their education.

This programme consists of 6 modules completed over 2 semesters. The full list of modules, and their corresponding credit value, is provided in [Section 5](#) of this handbook.

1.2 Programme Objectives

The overall objectives of this programme are:

1.	To provide graduates with the ability to advance their career by attaining a qualification which enables them to secure industry entry to junior level positions, or further advancement within their current sphere.
2.	To provide businesses with graduates who possess the requisite attributes to make a positive contribution to industry.
3.	To provide learners with an NQF level 8 major award that will provide them with the knowledge, skill and competence to set them on a progression pathway to further their education in the area of Data Analytics.
4.	To provide graduates with the foundation upon which they can further their studies at NQF Level 9 in Computing or one of many Computing-related disciplines (in Ireland or abroad) such as Computer Science, Computational Science, Information Systems, IT Management, Technology and Innovation Management, Information Security & Digital Forensics, Information Systems Processes, and others.
5.	To provide graduates with an environment conducive to personal development established through the use of peer learning, mentoring and formative feedback, thus allowing them to acquire new knowledge and the ability to identify their further learning needs and continuing professional development.

1.3 Academic Integrity

Academic integrity has been defined as¹:

“the commitment to, and demonstration of, honest and moral behaviour in an academic setting”²

It assumes that all interactions with higher education institutions are approached with honesty. This includes all documentation submitted to the institution for academic purposes. Responsibility for demonstrating academic integrity lies with the enrolled learner.

Academic integrity involves:

“Compliance with ethical and professional principles, standards and practices and a consistent system of values, that serves as guidance for making decisions and taking actions in education, research and scholarship”³

Maintaining academic integrity is critical to the reputation of higher education, and to the recognition of a graduate’s academic learning and resulting qualifications.

A useful video to help understand these issues is available [here](#).

Section 2: Programme Support

2.1 Faculty

Name	Role
Denis Cummins	Dean of Faculty
Dermot Glanville Cinthia Figueiredo	Faculty Coordinator

Other faculty member details are available on the Moodle page for this programme (available on your Moodle dashboard).

2.2 Moodle Page

CCT College uses Moodle as its open source Course Management System. Moodle is a fully web based application and provides a single point of access for all your learning needs at CCT College on and off campus.

Using Moodle you will be able to access programme resources and conduct various online exams and assignments in relation to your course materials.

You can also find information about academic dates in the semester schedule and make online assessment submissions. Moodle provides another online tool to supplement your in-class learning at CCT.

You can access the Moodle page for this programme via your Moodle dashboard.

2.3 Academic Calendar

The Academic Calendar for your programme is available [here](#).

NOTE: This is a provisional schedule which CCT College reserves the right to amend.

2.4 Student Handbook

CCT College's Student Handbook contains vital information for your studies.

You will also find useful information about CCT College's Quality Assurance policies such as the Reasonable Accommodation and Personal Mitigating Circumstances (PMC) policies.

This handbook is available [here](#).

Section 3: Programme structure

3.1 Units and credits

Module Title	Semester	ECTS Credits
1. Strategic Thinking	1 & 2	10
2. Statistical Techniques for Data Analysis	1	10
3. Data Preparation	1	10
4. Machine Learning	1	10
5. Data Visualisation Techniques	2	10
6. Machine Learning for Business	2	10
TOTAL Credits		60

3.2 Mode of learning

This is a full-time programme delivered via onsite, face-to-face lessons.

3.3 MIPLOs

Minimum Intended Programme Learning Outcomes (MIPOLs) identify what you will be expected to know, understand, and/or be able to demonstrate in order to be successful on the programme and achieve your qualification.

On successful completion of this Programme a learner will be able to:	
MIPLO 1	Demonstrate knowledge of the process of Data Analysis, including machine learning, searching, sorting, cleaning and visualising large-scale data programmatically. Demonstrate knowledge and understanding of areas of Mathematics, Statistics, Computer Science and Business Intelligence relevant to the subject domain.
MIPLO 2	Demonstrate knowledge of the theory and concepts of Data Analytics (language, tools and methodologies) and statistical theory. Demonstrate knowledge of the specialised and integrated nature of machine learning within the data analysis domain.
MIPLO 3	Demonstrate proficiency in utilising a broad based skill-set with specialised abilities that allow work on projects that incorporate multiple core areas.

	Research existing / legacy solutions to business centric data analysis, critically evaluate their success or failure and formulate an informed opinion on how this outcome has led to further innovation within this domain. Demonstrate the implementation of skills and use of tools in Statistics, data analysis and Business Intelligence; use these tools to effectively analyse and solve problems involving data sets. Work effectively and efficiently within given time and resource limits; design appropriate analysis solutions and implement and test solutions to agreed requirements standards, while being able to provide and present written reports.
MIPLO 4	Demonstrate the ability and understanding to design and develop data analysis strategies using appropriate languages and methods drawing on existing and emerging techniques. Demonstrate the quantitative and qualitative skills necessary for data analysis within a business context. Be able to work within a group and independently to interpret and report on analysis results. Have the capacity to use informed critical judgement for the prioritising of data feature selection requirements, developing appropriate planning, resource allocation, management, courses of action, and adopting suitable approaches to problem solving.
MIPLO 5	Research implement and test machine learning modelling projects within particular application domains while critically assessing specified system requirements. Apply data analysis skills and tools in a range of contexts to critically evaluate pre-existing knowledge and its use in current projects; be professionally responsible for decisions made while considering ethical and legal codes of practice within the subject domain. Be able to draw upon a substantial set of diagnostic, technical, interpersonal and planning skills as well as critical thinking, to implement both standard and innovative approaches to non-trivial analysis.
MIPLO 6	Be capable of working within a team or independently to identify and apply problem solving to high level issues within data analysis. Be able to work as a member of a multi-disciplinary and diverse team, drawing from personal initiative when required, including being able to take direction from supervisors, and feedback from others, to solve real-world and business problems, while interacting and communicating effectively with staff and colleagues at all levels of an organisation in an international context.
MIPLO7	Demonstrate an awareness of the existing limits of their own knowledge and have an understanding of the necessity of continuing professional development incorporating the ability of identifying knowledge gaps to lead to the processing, assimilating and applying of new knowledge and skills.
MIPLO8	Demonstrate the ability to identify and understand the wider ethical, social, cultural, political and business context that exists beyond the science of data analysis as well as the need for professional standards and ethics within the industry to maintain the integrity and independence in their professional judgement, subscribing to and promoting an ethical approach to the provision of

	data analysis and IT services, and to be fair and supportive of their colleagues and work environment.
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3.4 Resource Requirements

This is a Computing based programme and you will require a Windows-based machine equipped with at least 8GB of RAM and powered by an Intel Core i5 or AMD Ryzen 5 processor.

NOTE a Tablet will not be sufficient for the requirements of this programme.

You will use freely available, open source software.

Your lecturers will provide you with details about these resources, and how to access them, when you start your programme.

Section 4: Assessment

4.1 Assessment model

Semester 1:

Module Title	Continuous Assessment Weighting %	Supervised Project Weighting %
1. Strategic Thinking	0	100
2. Statistical Techniques for Data Analysis	100	0
3. Data Preparation	100	0
4. Machine Learning	100	0

Semester 2:

Module Title	Continuous Assessment Weighting %	Supervised Project Weighting %
1. Strategic Thinking	0	100
5. Data Visualisation Techniques	100	0
6. Machine Learning for Business	100	0

4.2 Assessment decisions

All programmes in CCT are assessed on the Percentage Grading Scheme (%). The pass mark is 40%.

You must achieve the minimum pass mark in each module. To find out the breakdown of assessment in each module, please refer to the module Moodle pages for more information on your module assessment.

Repeat module attempts are normally capped at a pass mark (40%). Students normally have four attempts to pass a module- the first sitting, and three repeat opportunities.

4.2.1 Dealing with late assignments

Developing effective workload and time management skills and practising punctuality are essential skills expected of a learner on a higher education programme. To reinforce the importance of this and to provide a disincentive for late submission of assessments CCT College operates a late submission policy where penalties are applied.

The following rules apply:

1. Assessments (with the exception of real-time assessments) will be accepted as late submissions up to and including 5 calendar days after the submission deadline.
2. Real-time assessments (in-class tests, Moodle quizzes, interviews, presentations, etc) cannot be late. Failure to attend a real-time assessment will result in a mark of 0% and will be considered as an assessment opportunity for the purpose of entitlement to repeats.
3. Except where otherwise indicated, CA is submitted online through Moodle.
4. Moodle will automatically keep track of all CA submissions. A Moodle message will notify the learner when their assignment has been submitted successfully.
5. Learners must submit the assignment for each subject by the due date indicated on the continuous assessment grid.
6. Where no extension has been granted and the assessment is submitted late, 10% of the mark for the assessment will be deducted for work submitted up to and including 5 calendar days late. A zero grade will be allocated to any assessment submitted later than 5 calendar days after the submission date.

4.2.2 Assessment feedback

Assessment feedback supports learning and is an essential part of your learning. Feedback allows you to see where you received a grade for assessment and where you can improve if necessary. It also allows you to appreciate their strengths and weaknesses in respect of your learning and development within a module.

You are entitled to assessment feedback in respect of each continuous assessment. Feedback may be automated, provided in written, audio, video format or verbally. Feedback can be individualised or provided to a small group or whole class.

4.2.3 Compensation

Pass by Compensation In certain circumstances, a learner awarded a mark between 35% and 39% may be entitled to pass by compensation. Pass by compensation is governed by the QQI Assessment and Standards, Revised 2013 and will only be applied in accordance with the requirements stated therein, or in a subsequent policy replacing that. In summary, Assessment and Standards allows for Pass by Compensation in the following circumstances:

Compensation can only be applied in the following circumstances:

- The student has been assessed for all stage modules and no module in the stage has been failed outright (below 35%).
- In the case of full-time students, the results are from the same sitting (session).
- The stage aggregate of credit-weighted excesses of percentage marks (over 40) is greater than or equal to twice the stage aggregate of credit-weighted deficits of marks (under 40) and the potentially compensatable results account for no more than one-third of the credit for the stage, i.e. 20 credits in a 60-credit stage or 10 credits in a 30-credit stage.

Compensation may be applied only to enable a student to pass a stage. At the award stage, a student who passes by compensation remains eligible for honours or other such classification higher than pass. Compensation does not change the result of the modules passed in that way. When reporting module passes by compensation (on the Diploma Supplement), the actual result

is returned, for example, 37% along with an indication that the module Pass has been granted by compensation.

In the programme assessment strategy and approved programme schedule, certain modules may be designated as not passable by compensation.

4.2.4 Calculating the overall qualification grade

The overall award classification is derived from the weighted average of all modules contributing to the award stage.

The following rules apply in line with the College's QA manual:

- The calculation of each overall module grade is the weighted total of all assessment components. Students are not required to pass all components but are required to attempt all components.
- In order to achieve the overall award, learners must pass at 40% in all modules except where pass by-compensation is applied.
- Pass-by compensation is applied to the full programme at the end of each stage by the Board of Examiners.

The following grade boundaries apply the overall award for Special Purpose Awards:

Percentage Point Average (PPA)	Award Classification
Greater than or equal to 70%	Distinction
60%- 69%	Merit Grade 1
50%-59%	Merit Grade 2
40%- 49%	Pass
35%-39%	Potentially compensable fail
Less than or equal to 34%	Outright fail

Section 5: Modules

Module 1: Strategic Thinking

Semester:	1 & 2
ECTS:	10

Introduction:

This 10 ECTS Capstone module floats across all semesters Strategic Thinking concepts are introduced purposefully as the module and programme develops. As this module is the capstone, and spans all semesters, the syllabus content usefully synchronises with the principal Problem Based Learning project and associated problem milestones, furthering the relevance of content to practice. The syllabus explores Problem Reduction Identification and Solution Mapping (PRISM), this then builds to project planning and team development specifically within this field.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Critically evaluate the relationship between information technology infrastructure and organisational competitive advantage.
2.	Critically analyse and select open source and proprietary software with a view to developing IT solutions for business and business-related IT problems.
3.	Utilise tools of strategic business analysis to evaluate the current macro and micro business environment with a view to formulating future action plans.
4.	Research emerging technologies and critically evaluate their impact on business and business information systems in general.
5.	Understand the relationship between data gathering / utilisation and business intelligence and its impact on industry policy.

Resources:

The CCT College Library reading list for this programme is available [here](#).

Module 2: Statistical Techniques for Data Analysis

Semester:	1
ECTS:	10

Introduction:

This 10 ECTS module forms the basis for Numerical methods, particularly those pertaining to statistics and probability which are central to the domain of data analytics. This module will equip the learner with statistical skills that are immediately applicable to basic data analysis tasks as well as serving as a foundation for more sophisticated techniques introduced in later modules.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Explore and evaluate datasets using descriptive statistical analyses.
2.	Formulate and test hypotheses using appropriate inferential statistical techniques and both evaluate and communicate the results effectively to peers and team members.
3.	Apply statistical analysis to appropriate datasets and critique the limitations of these models.
4.	Utilise current software tools and languages to produce and document result sets from existing data(e.g. Excel, R, Python).

Resources:

The CCT College Library reading list for this programme is available [here](#).

Module 3: Data Preparation

Semester:	1
ECTS:	10

Introduction:

This 10 ECTS module provides the learner with exposure to extensive exploratory data analysis and proper data preparation are a crucial first step in any data analysis process. The aim of this module is to provide the learner with an in-depth understanding of the rationale for data exploration and the methods used to explore data programmatically with a high level, low entry barrier language (e.g. python), The student also learns the importance of feature selection and dimensionality reduction and the bias-variance trade-off, the importance of the correct encoding of data and the usefulness of feature engineering as a means of representing complex functional relationships to machine learning models. By incorporating basic programming skills in a hands-on, practical integrated manner enables the learner's ability to program but also reinforces the inseparable nature of programming within the field of Data analytics. This integrated approach to practical programming skills is continued in accompanying modules.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Develop strategies, incorporating basic programming skills (input / output and basic data structures) for identifying and handling missing and out-of-range data.
2.	Programmatically implement graphical methods to identify issues within a data set (missing, out of range, dirty data)
3.	Engineer new features selection in data with the goal of improving the performance of machine learning models.
4.	Critically evaluate and implement suitable data-encoding techniques for a variety of machine learning algorithms.
5.	Collaboratively perform a critical analysis of a data set to optimise the data for a given problem space. Document the rationale behind the group's decisions to peers and stakeholders.

Resources:

The CCT College Library reading list for this programme is available [here](#).

Module 4: Machine Learning

Semester:	1
ECTS:	10

Introduction:

This 10 ECTS module provides the learner with Machine learning techniques that are an essential component of data analytics. This module builds on and draws from the Statistical techniques for Data Analysis and Data Preparation module to equip the learner with the ability to identify the fundamental nature of a data analytical problem and practical experience of the use of commonplace classification and regression approaches.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Implement Machine Learning Algorithms to solve analytical problems.
2.	Determine whether a given data analysis problem requires the use of supervised, semi-supervised or unsupervised learning methods. Develop and implement the chosen learning method.
3.	Develop a machine learning strategy for a given domain, communicate this strategy effectively to team members, peers and project stakeholders.
4.	Implement a range of classification and regression techniques and detail / document their suitability for a variety of problem domains.
5.	Critically evaluate and optimise the performance of Machine Learning models.

Resources:

The CCT College Library reading list for this programme is available [here](#).

Module 5: Data Visualisation Techniques

Semester:	2
ECTS:	10

Introduction:

This 10 ECTS Data Visualisation module is a key tool in the data analyst's toolbox, allowing the efficient and effective communication of vast quantities of data, offering rapid insights that would otherwise be difficult or impossible with numerical presentation. This module will provide the learner with the skills needed to present a variety of different types and volumes data in a manner that provides the maximum insight and understanding to the viewer. Allowing the learner to display directly the results of learning achieved in previous modules.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Discuss the concepts, techniques and processes underlying data visualisation.
2.	Critically evaluate visualisation approaches with respect to their suitability for different problem areas.
3.	Select appropriate data visualisation techniques for a given use case, data characteristics and multiple transmission media.
4.	Propose, design, develop, and implement an interactive data visualisation solution, for a given data set, detailing the rationale for approach and visualisation techniques choices made during development.
5.	Present a detailed visualisation of a data analysis to peers, team members and project stakeholders.

Resources:

The CCT College Library reading list for this programme is available [here](#).

Module 6: Machine Learning for Business

Semester:	2
ECTS:	10

Introduction:

This 10 ECTS module builds on the knowledge acquired in Machine Learning, focusing on the available machine learning algorithms widely integrated into commercial machine learning modelling. This module is designed to equip the learner with the skills necessary to tackle a wide range of unsupervised learning problems, such as cluster analysis and text analytics. Both of these techniques are widely used in the analysis of business data as they allow the enterprise to develop a deeper understanding of their customers. The module will also provide the learner with the necessary understanding to be able to perform modelling of temporal data, a type of data that is commonplace in the business domain.

Minimum intended module learning outcomes:

On completion of the module, the learner will be able to:

1.	Critically evaluate and implement appropriate clustering algorithms and interpret and document their results.
2.	Perform market basket analysis on representative transactional data. Explore the potential applications of these techniques outside their usual domain.
3.	Implement text categorisation, topic modelling and document summarisation on a range of representative texts.
4.	Apply modelling to time series data to facilitate business intelligence needs.

Resources:

The CCT College Library reading list for this programme is available [here](#).

Section 6: References

- 1 NAIN (2021) Academic Integrity: National Principles and Lexicon of Common Terms
- 2 Writing Center, University of North Carolina at Chapel Hill Academic Integrity
- 3 ENAI (2018) Glossary for Academic Integrity

Appendix A: Programme Schedule

Name of Provider:		CCT College Dublin												
Programme Title		Higher Diploma in Science in Data Analytics for Business												
Award Title		Higher Diploma in Science in Data Analytics for Business												
Stage Exit Award Title		None												
Modes of Delivery (FT/PT):		Full Time												
Teaching and learning modalities		Classroom instruction, supervised computer laboratories, independent learning, supervised individual and group project work												
Award Class	Award NFQ level	Award EQF Level	Stage (1, 2, 3, 4, ..., or Award Stage):		Stage NFQ Level ²		Stage EQF Level		Stage Credit (ECTS)		Date Effective		ISCED Subject code	
Major	8	7	Award		8		7		60		September 2020		0610	
Module Title		Semester	Module		Credit Number	Total Student Effort Module (hours)					Allocation of Marks (from the module assessment strategy)			
			Status ¹	NFQ Level where specified	Credit Units	Total Hours	Class (or equiv) Contact Hours	Directed e-learning	Hours of Independent Learning	Work-based learning effort	C.A. %	Supervised Project %	Proctored practical demonstration %	Proctored written exam %
					ECTS Credits									
1. Strategic Thinking		1 & 2	M	8	10	250	30	30	190	0	0	100	0	0
2. Statistical Techniques for Data Analysis		1	M	8	10	250	30	30	190	0	100	0	0	0
3. Data Preparation		1	M	8	10	250	30	30	190	0	100	0	0	0

1

4. Machine Learning	1	M	8	10	250	30	30	190	0	100	0	0	0
5. Data Visualisation Techniques	2	M	8	10	250	30	30	190	0	100	0	0	0
6. Machine Learning for Business	2	M	8	10	250	30	30	190	0	100	0	0	0
Special Regulations (Up to 280 characters)													
Mandatory (m) or elective (E)													